

```
1
2
3 2025-12-09 14:53          1-ping-pong/COMPONENT_TYPE_pingerC.h          Page 1
4
5
6
7 #define TYPEOF_COMPONENT_pingerC_VAR_requestMadeI  bool
8 #define TYPEOF_COMPONENT_pingerC_VAR_chosenAmountI  Amount
9 #define TYPEOF_COMPONENT_pingerC_VAR_amountI  Amount
10 #define TYPEOF_COMPONENT_pingerC_VAR_VALUEEXIST  bool
11 #define TYPEOF_COMPONENT_pingerC_VAR_PORT_pingerEmitterP_INDEX  byte
12
13 #define INITIALVALUE_COMPONENT_pingerC_VAR_requestMadeI  false
14 #define INITIALVALUE_COMPONENT_pingerC_VAR_chosenAmountI  Little
15
16 #define IC0_COMPONENT_pingerC_PORT_pingerEmitterP_ACTION_ping_WAITS \
17 !COMPONENT_pingerC_VAR_requestMadeI[0]
18
19 #define IC0_COMPONENT_pingerC_PORT_pingerEmitterP_ACTION_ping_PRE \
20 IC0_COMPONENT_pingerC_PORT_pingerEmitterP_ACTION_ping_WAITS
21
22 #define ICONTRACT_COMPONENT_pingerC_PORT_pingerEmitterP_ACTION_ping_PRE \
23 IC0_COMPONENT_pingerC_PORT_pingerEmitterP_ACTION_ping_PRE
24
25
26
```

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27
28 2025-12-09 14:53 1-ping-pong/COMPONENT_TYPE_pingerC_INSTANCE.pml Page 1
29
30
31 proctype instance_name(CompositeName, CompositeID, pingerC, CompInstanceID, Instance) () {
32
33
34   Component_i_roleData(pingerC, CompInstanceID, Instance);
35
36   TypeOf (COMPONENT_pingerC_VAR_amountI) COMPONENT_pingerC_VAR_amountI[1];
37   TypeOf (COMPONENT_pingerC_VAR_requestMadeI) COMPONENT_pingerC_VAR_requestMadeI[1]=InitialValue(
COMPONENT_pingerC_VAR_requestMadeI);
38   TypeOf (COMPONENT_pingerC_VAR_requestMadeI) PRE (COMPONENT_pingerC_VAR_requestMadeI[1]=InitialV
alue (COMPONENT_pingerC_VAR_requestMadeI);
39   TypeOf (COMPONENT_pingerC_VAR_chosenAmountI) COMPONENT_pingerC_VAR_chosenAmountI[1]=InitialValu
e (COMPONENT_pingerC_VAR_chosenAmountI);
40   TypeOf (COMPONENT_pingerC_VAR_chosenAmountI) PRE (COMPONENT_pingerC_VAR_chosenAmountI[1]=Initia
lValue (COMPONENT_pingerC_VAR_chosenAmountI);
41   TypeOf (COMPONENT_pingerC_VAR_VALUEEXIST) COMPONENT_pingerC_VAR_VALUEEXIST[1];
42   TypeOf (COMPONENT_pingerC_VAR_PORT_pingerEmitterP_INDEX) COMPONENT_pingerC_VAR_PORT_pingerEmi
terP_INDEX;
43
44   Start:
45   do
46     ::true->
47     if
48
49       ::atomic{ free->
50       #if (defined(CHECK_MONITOR) || defined(CHECK_LTL)) && defined(IMPOSE_ATOMIcity)
51         free = false;
52       #endif
53       select (COMPONENT_pingerC_VAR_PORT_pingerEmitterP_INDEX: 0 .. Component_i_Port_Num(Compositen
ame, CompositeID, pingerC, CompInstanceID, Instance, pingerEmitterP)-1) ->
54         COMPONENT_pingerC_VAR_amountI[0]=COMPONENT_pingerC_VAR_chosenAmountI[0];
55         if
56           :: refined_contract2 (CompositeName, CompositeID, pingerC, CompInstanceID, Instance, pingerEmi
tterP, ping) ()->
57           COMPONENT_pingerC_VAR_requestMadeI[0]=true;
58           COMPONENT_pingerC_VAR_VALUEEXIST[0]=false;
59           PRE (COMPONENT_pingerC_VAR_requestMadeI[0])=COMPONENT_pingerC_VAR_requestMadeI[0];
60
61           component_i_role_dataUpdate(CompositeName, CompositeID, pingerC, CompInstanceID, Instanc
e, pingerEmitterP, ping);
62           Component_i_channelName_action (CompositeName, CompositeID, pingerC, CompInstanceID, Instan
ce, pingerEmitterP, ping);
63           ::else-> progress0:goto Start
64         fi;
65       }
66       #if defined(CHECK_LTL)
67         pingerC_pingerEmitterP_ping_label_req:skip;
68       #endif
69       #if (defined(CHECK_MONITOR) || defined(CHECK_LTL)) && defined(IMPOSE_ATOMIcity)
70         free = true;
71       #endif
72
73
74     ::else->skip
75     fi;
76   od
77
78 }
79

```

```
80
81 2025-12-09 14:53          1-ping-pong/COMPONENT_TYPE_pongerC.h          Page 1
82
83
84 typedef PORT_pongerConsumerP_BUFFER {
85     bool slotOccupied;
86     mtype action;
87     int param_0;
88
89 }
90
91 #define TYPEOF_COMPONENT_pongerC_VAR_requestMadeO    bool
92 #define TYPEOF_COMPONENT_pongerC_VAR_chosenAmountO    Amount
93 #define TYPEOF_COMPONENT_pongerC_VAR_amountO          Amount
94 #define TYPEOF_COMPONENT_pongerC_VAR_VALUEEXIST       bool
95 #define TYPEOF_COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX    byte
96 #define TYPEOF_COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX    byte
97
98 #define INITIALVALUE_COMPONENT_pongerC_VAR_requestMadeO    false
99 #define INITIALVALUE_COMPONENT_pongerC_VAR_chosenAmountO    Little
100
101 #define IC0_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_WAITS \
102 COMPONENT_pongerC_VAR_requestMadeO[0] == true
103
104 #define IC0_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_ACCEPTS \
105 true
106 #define IC0_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_REJECT \
107 !(true)
108 #define IC0_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_PRE \
109 IC0_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_WAITS && \
110 IC0_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_ACCEPTS
111
112 #define ICONTRACT_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_PRE \
113 IC0_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_PRE
114 #define ICONTRACT_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_PRE_REJECT \
115 IC0_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_REJECT
116
117 #define FC0_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_REQUIRES \
118 COMPONENT_pongerC_VAR_PORT_pongerConsumerP_BUFFER[slot(CompositeName,CompositeID,pongerC,CompI
119 nstanceID,Instance,pongerConsumerP,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX,COMPONENT_
120 ngerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX)].param_0 == COMPONENT_pongerC_VAR_chosenAmou
121 ntO[0]
122
123 #define FCONTRACT_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_REQUIRES \
124 (FC0_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_REQUIRES)
125
```

```

126
127 2025-12-09 14:53 1-ping-pong/COMPONENT_TYPE_pongerC_INSTANCE.pml Page 1
128
129
130 proctype instance_name(CompositeName,CompositeID, pongerC,CompInstanceID,Instance) () {
131
132
133   Component_i_c_code(CompositeName,CompositeID,pongerC,CompInstanceID, Instance);
134
135   Component_i_roleData(pongerC,CompInstanceID, Instance);
136
137   TypeOf (COMPONENT_pongerC_VAR_amountO) COMPONENT_pongerC_VAR_amountO[1];
138   TypeOf (COMPONENT_pongerC_VAR_requestMadeO) COMPONENT_pongerC_VAR_requestMadeO[1]=InitialValue(
COMPONENT_pongerC_VAR_requestMadeO);
139   TypeOf (COMPONENT_pongerC_VAR_requestMadeO) PRE (COMPONENT_pongerC_VAR_requestMadeO[1])=InitialV
alue (COMPONENT_pongerC_VAR_requestMadeO);
140   TypeOf (COMPONENT_pongerC_VAR_chosenAmountO) COMPONENT_pongerC_VAR_chosenAmountO[1]=InitialValu
e (COMPONENT_pongerC_VAR_chosenAmountO);
141   TypeOf (COMPONENT_pongerC_VAR_chosenAmountO) PRE (COMPONENT_pongerC_VAR_chosenAmountO[1])=Initia
lValue (COMPONENT_pongerC_VAR_chosenAmountO);
142   TypeOf (COMPONENT_pongerC_VAR_VALUEEXIST) COMPONENT_pongerC_VAR_VALUEEXIST[1];
143   TypeOf (COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX) COMPONENT_pongerC_VAR_PORT_pongerCon
sumerP_INDEX;
144   TypeOf (COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX) COMPONENT_pongerC_VAR_PORT
_pongerConsumerP_CONNECTIONINDEX;
145
146   Component_i_port_buffer(pongerC,Instance,pongerConsumerP) COMPONENT_pongerC_VAR_PORT_pongerCo
nsumerP_BUFFER[Component_i_port_SUMCONNECTIONS(CompositeName,CompositeID,pongerC,CompInstanceI
D,Instance,pongerConsumerP)];
147
148   chan CHANNEL_COMPONENT_pongerC_VAR_PORT_pongerConsumerP[Component_i_Port_Num(CompositeName,Com
positeID,pongerC,CompInstanceID,Instance,pongerConsumerP) +1];
149
150
151   atomic {
152     COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX = 0;
153     CHANNEL_COMPONENT_pongerC_VAR_PORT_pongerConsumerP[Component_i_Port_Num(CompositeName,Composit
eID,pongerC,CompInstanceID,Instance,pongerConsumerP)] = component_i_blockedChannel(CompositeNa
me,CompositeID,pongerC,CompInstanceID,Instance);
154     do
155       ::COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX<Component_i_Port_Num(CompositeName,Composit
eID,pongerC,CompInstanceID,Instance,pongerConsumerP)->
156       CHANNEL_COMPONENT_pongerC_VAR_PORT_pongerConsumerP[COMPONENT_pongerC_VAR_PORT_pongerConsumerP
_INDEX] =Component_i_channelName(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pong
erConsumerP)[COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX];
157       COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX++;
158       ::else->COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX = 0; break
159     od ;
160
161     COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX = 0;
162     do
163       ::COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX<Component_i_Port_Num(CompositeName,Composit
eID,pongerC,CompInstanceID,Instance,pongerConsumerP)->
164       do
165         ::COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX<Component_i_port_NUMOFCONNECTIO
NS(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP)->
166         COMPONENT_pongerC_VAR_PORT_pongerConsumerP_BUFFER[slot(CompositeName,CompositeID,pongerC,CompI
nstanceID,Instance,pongerConsumerP,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX,COMPONENT_
pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX)].slotOccupied = false;
167         COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX++;
168         ::else->COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX=0; break
169       od;
170       COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX++
171       ::else->COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX=0;COMPONENT_pongerC_VAR_PORT_pongerCo
nsumerP_CONNECTIONINDEX=0;; break
172     od;
173
174   };
175   Start:
176   do
177     ::atomic {
178       COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX=0;
179       do
180         ::COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX<=Component_i_Port_Num(CompositeName,Com
positeID,pongerC,CompInstanceID,Instance,pongerConsumerP)-1 && (refined_contract2(CompositeNam

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```
180 e, CompositeID, pongerC, CompInstanceID, Instance, pongerConsumerP, ping) () -> break
181   :: COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX < Component_i_Port_Num (CompositeName, Comp
ositeID, pongerC, CompInstanceID, Instance, pongerConsumerP) -1 ->
182     if
183     :: true ->
184       COMPONENT_pongerC_VAR_VALUEEXIST[0] = true;
185     :: else
186
```

```

187
188 2025-12-09 14:53 1-ping-pong/COMPONENT_TYPE_pongerC_INSTANCE.pml Page 2
189
190
191         fi;          COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX++;
192     ::else->
193         if
194             ::COMPONENT_pongerC_VAR_VALUEEXIST[0]->break
195             ::else->break
196         fi;
197     od;
198     COMPONENT_pongerC_VAR_VALUEEXIST[0]=false;
199     COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX=0;
200     do
201         ::COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX<=Component_i_port_NUMOFCONNECTIONS(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP)-1 &&
202         COMPONENT_pongerC_VAR_PORT_pongerConsumerP_BUFFER[slot(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX)].slotOccupied-> break
203         ::COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX<Component_i_port_NUMOFCONNECTIONS(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP)-1->
204             COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX++;
205         ::COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX==Component_i_port_NUMOFCONNECTIONS(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP)-1-> break
206     od;
207 }
208 if
209     ::atomic{
210         CHANNEL_COMPONENT_pongerC_VAR_PORT_pongerConsumerP[(free -> COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX :Component_i_Port_Num(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP))] ? eval(ping) ,COMPONENT_pongerC_VAR_amountO[0]->
211         #if (defined(CHECK_MONITOR) || defined(CHECK_LTL)) && defined(IMPOSE_ATOMICTY)
212             free = false;
213         #endif
214         COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX=0;
215         do
216             ::COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX<=Component_i_port_NUMOFCONNECTIONS(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP)-1 &&
217             !COMPONENT_pongerC_VAR_PORT_pongerConsumerP_BUFFER[slot(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX)].slotOccupied-> break
218             ::COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX<Component_i_port_NUMOFCONNECTIONS(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP)-1 &&
219             COMPONENT_pongerC_VAR_PORT_pongerConsumerP_BUFFER[slot(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX)].slotOccupied->
220                 COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX++;
221             ::COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX==Component_i_port_NUMOFCONNECTIONS(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP)-1-> break
222         od;
223         #ifdef CHECK_BUFFEROVERFLOW
224             assert (COMPONENT_pongerC_VAR_PORT_pongerConsumerP_BUFFER[slot(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX)].slotOccupied==false);
225         #endif
226         COMPONENT_pongerC_VAR_PORT_pongerConsumerP_BUFFER[slot(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX)].action=ping
227     ;
228     COMPONENT_pongerC_VAR_PORT_pongerConsumerP_BUFFER[slot(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX)].param_0=COMPONENT_pongerC_VAR_amountO[0]
229     ;
230     COMPONENT_pongerC_VAR_PORT_pongerConsumerP_BUFFER[slot(CompositeName,CompositeID,pongerC,CompInstanceID,Instance,pongerConsumerP,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX)].slotOccupied = true;
231 }
232 progress3:skip;
233 #if (defined(CHECK_MONITOR) || defined(CHECK_LTL)) && defined(IMPOSE_ATOMICTY)
234     free = true;
235 #endif

```

```
239
240     ::atomic{
241         COMPONENT_pongerC_VAR_PORT_pongerConsumerP_BUFFER[slot (CompositeName, CompositeID, pongerC, CompInstanceID, Instance, pongerConsumerP, COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX, COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX)].slotOccupied &&
242 COMPONENT_pongerC_VAR_PORT_pongerConsumerP_BUFFER[slot (CompositeName, CompositeID, pongerC, CompInstanceID, Instance, pongerConsumerP, COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX, COMPONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX)].action == ping && refined_contract2 (CompositeName, CompositeID, pongerC, CompInstanceID, Instance, pongerConsumerP, ping) () && free->
243 #if (defined(CHECK_MONITOR) || defined(CHECK_LTL)) && defined(IMPOSE_ATOMICITY)
244     free = false;
245 #endif
246 if
247
```

```
248
249 2025-12-09 14:53 1-ping-pong/COMPONENT_TYPE_pongerC_INSTANCE.pml Page 3
250
251
252      ::FC0_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_REQUIRES->skip;
253      COMPONENT_pongerC_VAR_requestMadeO[0]=false;
254      COMPONENT_pongerC_VAR_VALUEEXIST[0]=false;
255      PRE (COMPONENT_pongerC_VAR_requestMadeO[0])=COMPONENT_pongerC_VAR_requestMadeO[
0];
256
257      ::else->skip;
258      #ifdef CHECK_CHAOS
259      assert(false)
260      #endif
261      fi;
262      component_i_role_dataUpdate(CompositeName, CompositeID,pongerC,CompInstanceID, Instance,
pongerConsumerP,ping);
263      COMPONENT_pongerC_VAR_PORT_pongerConsumerP_BUFFER[slot(CompositeName,CompositeID,pongerC
,CompInstanceID,Instance,pongerConsumerP,COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX,COMP
ONENT_pongerC_VAR_PORT_pongerConsumerP_CONNECTIONINDEX)].slotOccupied=false
264      ;
265      }
266      #if defined(CHECK_LTL)
267      pongerC_pongerConsumerP_ping_label_req:skip;
268      #endif
269      #if (defined(CHECK_MONITOR) || defined(CHECK_LTL)) && defined(IMPOSE_ATOMICITY)
270      free = true;
271      #endif
272
273      ::else->skip
274      fi;
275      od
276
277  }
278
```


279

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281

282

283

284

285

286

```
287
288 2025-12-09 14:53 1-ping-pong/COMPONENT_TYPE_together_INSTANCE.pml Page 1
289
290
291 proctype instance_name(CompositeName, CompositeID, together, CompInstanceID, Instance) () {
292
293   run instance_name(CompositeName, CompositeID, pingerC, pi, 0) ();
294   run instance_name(CompositeName, CompositeID, pongerC, po, 0) ();
295
296
297
298 }
299
```

300

301 2025-12-09 14:53 1-ping-pong/CONNECTOR_TYPE_linkEm_link_0_ROLE_pingerR.h Page 1

302

303

304 #define ICONTRACT_CONNECTOR_linkEm_link_0_ROLE_pingerR_PORTVAR_pingerEmitterVarP_ACTION_ping_P
RE \

305 true

306

307

308 2025-12-09 14:53 1-ping-pong/CONNECTOR_TYPE_linkEm_link_0_ROLE_pongerR.h Page 1

309

310

311 #define ICONTRACT_CONNECTOR_linkEm_link_0_ROLE_pongerR_PORTVAR_pongerConsumerVarP_ACTION_ping_
PRE \

312 true

313 #define TYPEOF_CONNECTOR_linkEm_ROLE_pongerR_VAR_PORTVAR_pongerConsumerVarP_INDEX byte

314

315

316 2025-12-09 14:53

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317

318

319 #ifndef Action

320 #define Action mtype

321 mtype = {

322 ping}

323 #endif

324

```
325
326 2025-12-09 14:53          1-ping-pong/TYPE_Amount.h          Page 1
327
328
329 #ifndef Amount
330 #define Amount  mtype
331 mtype = {Much,Average,Little,None}
332 #endif
333
```

334

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336

337

338 #ifndef ID

339 #define ID byte

340 #endif

341

```
342
343 2025-12-09 14:53          1-ping-pong/Xcd_PACKAGE.h          Page 1
344
345
346 #ifndef PACKAGE
347 #define PACKAGE 1
348
349 // #include "TYPE_ID.h"
350 // #include "TYPE_Action.h"
351 // #include "TYPE_Amount.h"
352
353 /*****Constants*****/
354 #define RESULT 0
355 #define EXCEPTION 1
356 #define bit_MIN 0
357 #define bit_MAX 1
358 #define bool_MIN 0
359 #define bool_MAX 1
360 #define byte_MIN 0
361 #define byte_MAX 255
362 #define mtype_MIN 0
363 #define mtype_MAX 255
364
365 /*****Macros for Conditional Expressions*****/
366 /// macro for returning AND of expressions
367 #define AND(a, b) ((a) && (b))
368 /// macro for returning OR of expressions
369 #define OR(a, b) ((a) || (b))
370 /// macro for returning IMPLIES
371 #define IMPLIES(a, b) (!(a) || (b))
372
373 /*****Macro for naming convention*****/
374 #define Component_i_smt(component, i, smt) Component_i_2_smt(component, i, smt)
375 #define Component_i_2_smt(component, i, smt) COMPONENT_##component##_##i ## smt
376
377 #define Component_i_smt_ins(component, compIns, i, smt) Component_i_2_smt_ins(component, compIns,
i, smt)
378 #define Component_i_2_smt_ins(component, compIns, i, smt) COMPONENT_##component##_##compIns##_##
i ## smt
379
380 #define SubComponent_i_smt(compositeName, compositeIndex, component, compIns, i, smt) \
381     SubComponent_i_2_smt(compositeName, compositeIndex, component, compIns, i, smt)
382 #define SubComponent_i_2_smt(compositeName, compositeIndex, component, compIns, i, smt) compositeN
ame##_##compositeIndex##_COMPONENT_##component##_##compIns##_##i ## smt
383
384 #define SubConnector_i_smt(compositeName, compositeIndex, connector, connIns_i, smt) \
385     SubConnector_i_2_smt(compositeName, compositeIndex, connector, connIns_i, smt)
386 #define SubConnector_i_2_smt(compositeName, compositeIndex, connector, connIns_i, smt) compositeNa
me##_##compositeIndex##_CONNECTOR_##connector##_##connIns_i ## smt
387
388 #define Component_i_smt_pre_post(pre, component, i, smt) Component_i_2_pre_post(pre, component,
i, smt)
389 #define Component_i_2_pre_post(pre, component, i, smt) pre##COMPONENT_##component##_##i ## smt
390 /*****Macros for VARIABLE Operations*****/
391 // macro for returning process names for component instances
392 #define instance_name(CompositeName, CompositeID, SubCompType, SubCompIns, Instance) \
393     instance_name2(CompositeName, CompositeID, SubCompType, SubCompIns, Instance)
394 #define instance_name2(CompositeName, CompositeID, SubCompType, SubCompIns, Instance) \
395     COMPONENT_TYPE_##CompositeName##_##CompositeID##_SUBCOMPONENT_## SubCompType##_
##SubCompIns##_##Instance
396 // macro for returning the minimum value of a type
397 #define Min_of_Type(Type) Type##_MIN
398 // macro for returning the maximum value of a type
399 #define Max_of_Type(Type) Type##_MAX
400 // macro for returning the minimum value for a variable
401 #define Min_of_Var(Var) Min_of_Var2(TYPEOF_##Var)
402
```



```

403
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405
406
407 #define Min_of_Var2(Type) Min_of_Type(Type)
408 // macro for returning the minimum value for a variable
409 #define Max_of_Var(Var) Max_of_Var2(TYPEOF_##Var)
410 #define Max_of_Var2(Type) Max_of_Type(Type)
411 //macro for returning the variable name that holds a result
412 #define Result_var(ComponentType, port, method) COMPONENT_##ComponentType ##_VAR_PORT_##port #
  _METHOD_##method##_RESULT
413 //macro for returning the variable name that holds a pre value of a variable
414 #define PRE(X) pre_##X
415 #define TEMP(X) temp_##X
416 //macro for returning the variable name that holds a post value of a variable
417 #define POST(X) post_##X
418 //macro for returning the constant for a variable type
419 #define TypeOf(X) TYPEOF_##X
420 //macro for returning the constant for an initial value of a variable
421 #define InitialValue(X) INITIALVALUE_##X
422 //macro for returning the constant for an initial pre value of a variable
423 #define InitialValue_pre(X) InitialValue_pre_##X
424 //macro for returning variable name
425 #define Variable_Name(compType, variable) COMPONENT_##compType##_VAR_##variable
426 //macro for returning refined contract id
427 #define refined_contract_request(compositeName, compositeIndex, comptype, compIns, Instance, p
  ortName,action) \
428 SubComponent_i_smt (compositeName,compositeIndex,comptype,compIns, Instance, _PORT_##portName##
  _##ACTION_##action##_REFINED_IContract_REQUEST)
429
430 #define refined_contract_response(compositeName, compositeIndex, comptype, compIns, Instance,
  portName,action) \
431 SubComponent_i_smt (compositeName,compositeIndex,comptype,compIns, Instance, _PORT_##portName##
  _##ACTION_##action##_REFINED_IContract_RESPONSE)
432
433 #define refined_contract2(compositeName, compositeIndex, comptype, compIns,Instance, portName,
  action) \
434 SubComponent_i_smt (compositeName,compositeIndex,comptype,compIns, Instance, _PORT_##portName##
  _##ACTION_##action##_REFINED_IContract)
435
436 #define refined_contract_reject(compositeName, compositeIndex, comptype, compIns,Instance, por
  tName,action) \
437 SubComponent_i_smt (compositeName,compositeIndex,comptype,compIns, Instance, _PORT_##portName##
  _##ACTION_##action##_REFINED_IContract_REJECT)
438
439 #define port_index(CompositeName, CompositeID,compType,CompInsID,Instance, portName, portID) \
440 c_expr{[ValueOfProcessLocalVar(CompositeName, CompositeID, compType, CompInsID,Instance, port
  ID)]};
441
442 //macro for returning the port size array
443 #define Component_i_port_j_Size(compType, Instance, portName) Component_i_smt (compType,Instan
  ce,portName##_SIZE)
444
445 /*****Macros for Component Instance Specification*****/
446
447 //COMPONENT PARAMETER
448 ///Macro for instance PARAMETER
449 #define Component_i_Param_N(compositeName,compositeID,component,compIns,Instance,N) \
450 SubComponent_i_smt (compositeName,compositeID,component,compIns,Instance,_PARAM_##N)
451
452 #define Connector_i_Param_N(compositeName,compositeID,connector,connIns_i,N) \
453 SubConnector_i_smt (compositeName,compositeID,connector,connIns_i,_PARAM_##N)
454
455
456 //COMPONENT PORT
457 ///Macro for number of instances for a PORT
458 #define Component_i_Port_Num(compositeName, compositeIndex, component,compIns,Instance, pname)
  \
459 SubComponent_i_smt (compositeName,compositeIndex,component,compIns,Instance,_PORT_#
  #pname##_NUM)
460 ///Macro for number of connected PORTS to CONSUMER/PROVIDED PORT instance
461 #define Component_i_port_j_CONNECTIONS(compositeName,compositeIndex,component,compIns,Instance
  ,j, port) Component_i_port_j_CONNECTIONS_2(compositeName,compositeIndex,component,compIns,Inst
  ance,j, port)
462 #define Component_i_port_j_CONNECTIONS_2(compositeName, compositeIndex, component,compIns,Inst

```

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462 ance, j, port) \

463

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466
467
468 SubComponent_i_smt (compositeName, compositeIndex, component, compIns, Instance, _PORT_##port##_##j#
  _CONNECTIONS)
469 ///Macro for number of connected PORTS to ALL instances of a CONSUMER/PROVIDED PORT
470 #define Component_i_port_SUMCONNECTIONS (compositeName, compositeIndex, component, compIns, Inst
  ance, port) \
471 SubComponent_i_smt (compositeName, compositeIndex, component, compIns, Instance, _PORT_##port##_SUMC
  ONNECTIONS)
472 ///Macro for number of connected PORTS to CONSUMER/PROVIDED PORT instance
473 #define Component_i_port_NUMOFCONNECTIONS (compositeName, compositeIndex, component, compIns, In
  stance, port) \
474 SubComponent_i_smt (compositeName, compositeIndex, component, compIns, Instance, _PORT_##port##_NUMO
  FCONNECTIONS)
475 ///macro for a name of the array that holds the offset of the CONSUMER/PROVIDED PORTS
476 #define Component_i_port_offsetArray (compositeName, compositeIndex, component, compIns, i, pname
  ) \
477 SubComponent_i_smt (compositeName, compositeIndex, component, compIns, i, _PORT_##pname##_OFFSE
  T)
478
479 ///macro for buffer of the CONSUMER/PROVIDED PORTS
480 #define Component_i_port_buffer (component, Instance, port) PORT_##port##_BUFFER
481 ///macro for a name of the array that holds the number of CONNECTIONS for the instances of a
  PORT
482 #define Component_i_port_connectionArray (component, Instance, pname) Component_i_smt (component
  , Instance, _PORT_##pname##_CONNECTS)
483 ///macro for an array slot that stores the number of CONNECTIONS for a PORT INSTANCE
484 #define port_connection_size (CompositeName, CompositeID, Component, CompInsID, Instance, pname,
  pnum) \
485 c_expr {Component_i_port_connectionArray (Component, Instance, pname) [ValueOfProc
  essLocalVar (CompositeName, CompositeID, Component, CompInsID, Instance, pnum)]}
486 ///macro for an array index that stores the BUFFER value received from a specific CONNECTION
  of a PORT INSTANCE
487 #define slot (CompositeName, CompositeID, Component, CompInsID, Instance, pname, pnum, conn) \
488 c_expr {Component_i_port_offsetArray (CompositeName, CompositeID, Component, CompInsID,
  Instance, pname) [ValueOfProcessLocalVar (CompositeName, CompositeID, Component, CompInsID, Insta
  nce, pnum)]} + conn
489 #define slot2 (CompositeName, CompositeID, Component, CompInsID, Instance, pname, pnum, conn) \
490 c_expr {Component_i_port_offsetArray (CompositeName, CompositeID, Component, CompInsID,
  Instance, pname) [ValueOfProcessLocalVar (CompositeName, CompositeID, Component, CompInsID, Insta
  nce, pnum)]} % 1 + conn
491 ///macro for returning the statement that enable the use of a process variable in a c_expr
492 #define ValueOfProcessLocalVar (CompositeName, CompositeID, Component, CompInstance, Instance,
  var) \
493 PCOMPONENT_TYPE_##CompositeName##_##CompositeID##_SUBCOMPONENT_## Component##_##CompIns
  tance##_##Instance-> var
494
495 ///macro for the name of the array which stores the INDEX of a CHANNEL that the INSTANCES of
  a PORT uses
496 #define Component_i_portChannelIndexArray (compType, Instance, pname) Component_i_smt (component
  , Instance, _PORT_##pname##_CHINDEXARRAY)
497 ///macros for the INDEX of a CHANNEL that an INSTANCE of a PORT uses
498 #define Component_i_portChannelIndex (CompositeName, CompositeID, compType, Instance, pname, pnu
  m) \
499 c_expr {Component_i_portChannelIndexArray (compType, Instance, pname) [ValueOfProce
  ssLocalVar (CompositeName, CompositeID, compType, Instance, pnum)]}
500 #define Component_i_portChannelIndex_CONS (compType, Instance, pname, consVal) \
501 c_expr {Component_i_portChannelIndexArray (compType, Instance, pname) [consVal]}
502 ///macro for the INDEX value of a CHANNEL that a COMPONENT PORT uses
503 #define Component_i_channelIndexValue (compType, cInstance, pname, pInstance) \
504 Component_i_smt (compType, cInstance, _PORT_##pname##_##pInstance##_CHINDE
  X)
505
506 ///macros for the name of the CHANNELS that a COMPONENT PORT uses
507 #define Component_i_channelName (compositeName, compositeIndex, compType, compIns, cInstance, pn
  ame) \
508 Component_i_smt_CH_ins (compositeName, compositeIndex, CHANNEL_, compType, compIns, cInsta
  nce, _PORT_##pname)
509
510 #define Component_i_channelName_action (compositeName, compositeIndex, compType, compInstanceID,
  cInstance, pname, action) \
511 Component_i_smt_CH_ins (compositeName, compositeIndex, CHANNEL_, compType, compInstanceID, cInsta
  nce, _PORT_##pname##_ACTION_##action)

```

```
512
513 #define Component_i_channelName_action_req(compositeName, compositeIndex, compType, compInstance
ID, cInstance, pname, action) \
514 Component_i_smt_CH_ins(compositeName, compositeIndex, CHANNELREQ_, compType, compInstanceID, cIn
stance, _PORT_##pname##_ACTION_##action)
515
516 #define Component_i_channelName_action_res(compositeName, compositeIndex, compType, compInstanc
eID, cInstance, pname, action) \
517 Component_i_smt_CH_ins(compositeName, compositeIndex, CHANNELRES_, compType, compInstanceID, cIn
stance, _PORT_##pname##_ACTION_##action)
518
519 #define Component_i_channelName_req(compositeName, compositeIndex, compType, compInstanceID, c
Instance, pname) \
520 Component_i_smt_CH_ins(compositeName, compositeIndex, CHANNELREQ_, compType, compIns
tanceID, cInstance, _PORT_##pname)
521
522 #define Component_i_channelName_res(compositeName, compositeIndex, compType, compInstanceID, cI
nstance, pname) \
523 Component_i_smt_CH_ins(compositeName, compositeIndex, CHANNELRES_, compType, compIns
tanceID, cInstance, _PORT_##pname)
524
```

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525
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527
528
529
530
531 #define Component_i_smt_CH_ins(compositeName, compositeIndex, chpre, compType, compInstanceID, i
, smt) \
532     Component_i_2_smt_CH_ins(compositeName, compositeIndex, chpre, compType, compIns
tanceID, i, smt)
533 #define Component_i_2_smt_CH_ins(compositeName, compositeIndex, chpre, component, compInstanceID,
i, smt) \
534     chpre##compositeName##_##compositeIndex##_##COMPONENT_##component##_##compIns
tanceID##_## i ## smt
535
536
537 #define Component_i_smt_CH(compositeName, compositeIndex, chpre, compType, i, smt) \
538     Component_i_2_smt_CH(compositeName, compositeIndex, chpre, compType, i, smt)
539 #define Component_i_2_smt_CH(compositeName, compositeIndex, chpre, component, i, smt) \
540     chpre##compositeName##_##compositeIndex##_##COMPONENT_##component##_## i ## s
mt
541
542 //macro for the name of c_code block
543 #define Component_i_c_code(compositeName, compositeIndex, compType, compIns, Instance) SubCompon
ent_i_smt (compositeName, compositeIndex, compType, compIns, Instance, _ARRAY_INITIALISATIONS) ()
544 //macro for the name of the role data block
545 #define Component_i_roleData(compType, compInstance, Instance) Component_i_smt_ins (compType, com
pInstance, Instance, _ROLE_DATA) ()
546
547 #define component_i_blockedChannel(compositeName, compositeID, compType, compInstance, Instanc
e) \
548 SubComponent_i_smt (compositeName, compositeID, compType, compInstance, Instance, _BLOCKEDCHANNEL)
549
550 #define component_i_role_dataUpdate(compositeName, compositeID, compType, compIns, Instance, po
rt, action) \
551 SubComponent_i_smt (compositeName, compositeID, compType, compIns, Instance, _PORT_##port##_ACTION_#
#action##_ROLE_DATAUPDATE) ()
552
553 #define component_i_role_dataUpdate_request(compositeName, compositeID, compType, compIns, Inst
ance, port, action) \
554 SubComponent_i_smt (compositeName, compositeID, compType, compIns, Instance, _PORT_##port##_ACTION_#
#action##_ROLE_DATAUPDATE_REQUEST) ()
555
556 #define component_i_role_dataUpdate_response(compositeName, compositeID, compType, compIns, Ins
tance, port, action) \
557 SubComponent_i_smt (compositeName, compositeID, compType, compIns, Instance, _PORT_##port##_ACTION_#
#action##_ROLE_DATAUPDATE_RESPONSE) ()
558
559 #endif
560
561
562
563
```

```
564
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566
567
568 // #define IMPOSE_ATOMICITY
569 // #define CHECK_LTL
570 // #define CHECK_MONITOR
571 bit free = true;
572 #define CHECK_RACE
573 #define CHECK_CHAOS
574 // #define CHECK_BUFFER_OVERFLOW
575 #ifndef COMPOSITE
576 #include "together_COMPOSITE.h"
577 #define COMPOSITE
578 #endif
579 #define CompositeName together
580 #define CompositeID 0
581 #define CompInstanceID 0
582 #define Instance 0
583 #include "COMPONENT_TYPE_together_INSTANCE.pml"
584 init{
585   run instance_name(CompositeName, CompositeID, together, CompInstanceID, Instance) ()
586 }
587
```

```
588
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590
591
592 #ifndef TYPE_Action
593 #define TYPE_Action
594 #include "TYPE_Action.h"
595 #endif
596 #ifndef TYPE_Exceptions
597 #define TYPE_Exceptions
598 #include "TYPE_EXCEPTIONS.h"
599 #endif
600 #ifndef TYPE_Amount
601 #define TYPE_Amount
602 #include "TYPE_Amount.h"
603 #endif
604 #ifndef TYPE_ID
605 #define TYPE_ID
606 #include "TYPE_ID.h"
607 #endif
608 #include "XcD_PACKAGE.h"
609 #define CompositeName together
610 #ifndef CompositeID
611 #define CompositeID 0
612 #endif
613 #ifndef COMPONENT_TYPE_pingerC
614 #define COMPONENT_TYPE_pingerC
615 #include "COMPONENT_TYPE_pingerC.h"
616 #endif
617
618 #ifndef COMPONENT_TYPE_pongerC
619 #define COMPONENT_TYPE_pongerC
620 #include "COMPONENT_TYPE_pongerC.h"
621 #endif
622
623
624 #ifndef CONNECTOR_TYPE_linkEm_link_0_ROLE_pingerR
625 #define CONNECTOR_TYPE_linkEm_link_0_ROLE_pingerR
626 #include "CONNECTOR_TYPE_linkEm_link_0_ROLE_pingerR.h"
627 #endif
628 #ifndef CONNECTOR_TYPE_linkEm_link_0_ROLE_pongerR
629 #define CONNECTOR_TYPE_linkEm_link_0_ROLE_pongerR
630 #include "CONNECTOR_TYPE_linkEm_link_0_ROLE_pongerR.h"
631 #endif
632
633
634
635 chan CHANNEL_together_0_COMPONENT_pongerC_po_0_PORT_pongerConsumerP[1] = [1] of {mtype,int};
636
637 #define CONNECTOR_linkEm_link_0_ROLE_pingerR_VAR_amountI COMPONENT_pingerC_VAR_amountI[0]
638 #define CONNECTOR_linkEm_link_0_ROLE_pingerR_VAR_PORT_pingerEmitterP_INDEX COMPONENT_pingerC_V
639 AR_PORT_pingerEmitterP_INDEX
640
641
642
643 #define CONNECTOR_linkEm_link_0_ROLE_pongerR_VAR_amountO COMPONENT_pongerC_VAR_amountO[0]
644 #define CONNECTOR_linkEm_link_0_ROLE_pongerR_VAR_PORT_pongerConsumerP_INDEX COMPONENT_pongerC_
645 VAR_PORT_pongerConsumerP_INDEX
646
647
648
```

```

649
650 2025-12-09 14:53          1-ping-pong/together_COMPOSITE.h          Page 2
651
652
653
654 //Component Instance : pingerC_pi_0
655 #define Instance 0
656 #define CompInstanceID pi
657 chan component_i_blockedChannel(CompositeName, CompositeID,pingerC,CompInstanceID,Instance)= [
658 0] of {mtype};
659 #define CHANNEL_together_0_COMPONENT_pingerC_pi_0_PORT_pingerEmitterP_ACTION_ping \
660 if\
661 ::COMPONENT_pingerC_VAR_PORT_pingerEmitterP_INDEX==0->\
662 CHANNEL_together_0_COMPONENT_pongerC_po_0_PORT_pongerConsumerP[0]!ping,COMPONENT_pingerC_VAR_a
663 mountI[0] \
664 fi
665
666 #define together_0_COMPONENT_pingerC_pi_0_PORT_pingerEmitterP_NUM 1
667
668 #define together_0_COMPONENT_pingerC_pi_0_PORT_pingerEmitterP_ACTION_ping_REFINED_ICONTRACT()
669 \
670 ICONTRACT_COMPONENT_pingerC_PORT_pingerEmitterP_ACTION_ping_PRE && (COMPONENT_pingerC_VAR_PORT
671 _pingerEmitterP_INDEX==0->ICONTRACT_CONNECTOR_linkEm_link_0_ROLE_pingerR_PORTVAR_pingerEmitter
672 VarP_ACTION_ping_PRE: true)
673
674
675 #define together_0_COMPONENT_pingerC_pi_0_PORT_pingerEmitterP_0_ACTION_ping_ROLE_DATAUPDATE()
676 \
677 true
678
679 #define together_0_COMPONENT_pingerC_pi_0_PORT_pingerEmitterP_ACTION_ping_ROLE_DATAUPDATE() \
680 if\
681 ::COMPONENT_pingerC_VAR_PORT_pingerEmitterP_INDEX == 0 -> \
682 together_0_COMPONENT_pingerC_pi_0_PORT_pingerEmitterP_0_ACTION_ping_ROLE_DATAUPDATE() \
683 fi;
684
685 #define COMPONENT_pingerC_pi_0_ROLE_DATA() \
686 true
687
688 #define XX together
689 #include "COMPONENT_TYPE_pingerC_INSTANCE.pml"
690 #undef CompInstanceID
691 #undef Instance
692 #undef XX
693
694 //Component Instance : pongerC_po_0
695 #define Instance 0
696 #define CompInstanceID po
697 chan component_i_blockedChannel(CompositeName, CompositeID,pongerC,CompInstanceID,Instance)= [
698 0] of {mtype};
699
700 #define together_0_COMPONENT_pongerC_po_0_PORT_pongerConsumerP_NUM 1
701
702 #define together_0_COMPONENT_pongerC_po_0_PORT_pongerConsumerP_ACTION_ping_REFINED_ICONTRACT()
703 \
704 ICONTRACT_COMPONENT_pongerC_PORT_pongerConsumerP_ACTION_ping_PRE && (COMPONENT_pongerC_VAR_POR
705 T_pongerConsumerP_INDEX==0->ICONTRACT_CONNECTOR_linkEm_link_0_ROLE_pongerR_PORTVAR_pongerConsu
706 merVarP_ACTION_ping_PRE: true)
707
708 #define together_0_COMPONENT_pongerC_po_0_PORT_pongerConsumerP_0_CONNECTIONS 1
709
710 #define together_0_COMPONENT_pongerC_po_0_PORT_pongerConsumerP_SUMCONNECTIONS \

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710
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712
713
714 together_0_COMPONENT_pongerC_po_0_PORT_pongerConsumerP_0_CONNECTIONS
715 #define together_0_COMPONENT_pongerC_po_0_PORT_pongerConsumerP_NUMOFCONNECTIONS \
716 (\
717 COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX==0->together_0_COMPONENT_pongerC_po_0_PORT_po
718 ngerConsumerP_0_CONNECTIONS :0)\
719
720
721
722 #define together_0_COMPONENT_pongerC_po_0_PORT_pongerConsumerP_0_ACTION_ping_ROLE_DATAUPDATE()
723 \
724 true
725
726 #define together_0_COMPONENT_pongerC_po_0_PORT_pongerConsumerP_ACTION_ping_ROLE_DATAUPDATE() \
727 if\
728 ::COMPONENT_pongerC_VAR_PORT_pongerConsumerP_INDEX == 0 -> \
729     together_0_COMPONENT_pongerC_po_0_PORT_pongerConsumerP_0_ACTION_ping_ROLE_DATAUPDATE() \
730 fi;
731
732 #define COMPONENT_pongerC_po_0_ROLE_DATA() \
733 true
734
735 c_code {
736 int Component_i_port_offsetArray (CompositeName, CompositeID, pongerC, CompInstanceID, 0, pongerCons
737 umerP) [Component_i_Port_Num (CompositeName, CompositeID, pongerC, CompInstanceID, 0, pongerConsumerP
738 )];
739 }
740
741 #define together_0_COMPONENT_pongerC_po_0_ARRAY_INITIALISATIONS() \
742 c_code { \
743 Component_i_port_offsetArray (CompositeName, CompositeID, pongerC, CompInstanceID, 0, pongerConsumer
744 P) [0] = 0;\
745 }
746
747 #define XX together
748 #include "COMPONENT_TYPE_pongerC_INSTANCE.pml"
749 #undef CompInstanceID
750 #undef Instance
751 #undef XX
752
753 end
```

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end